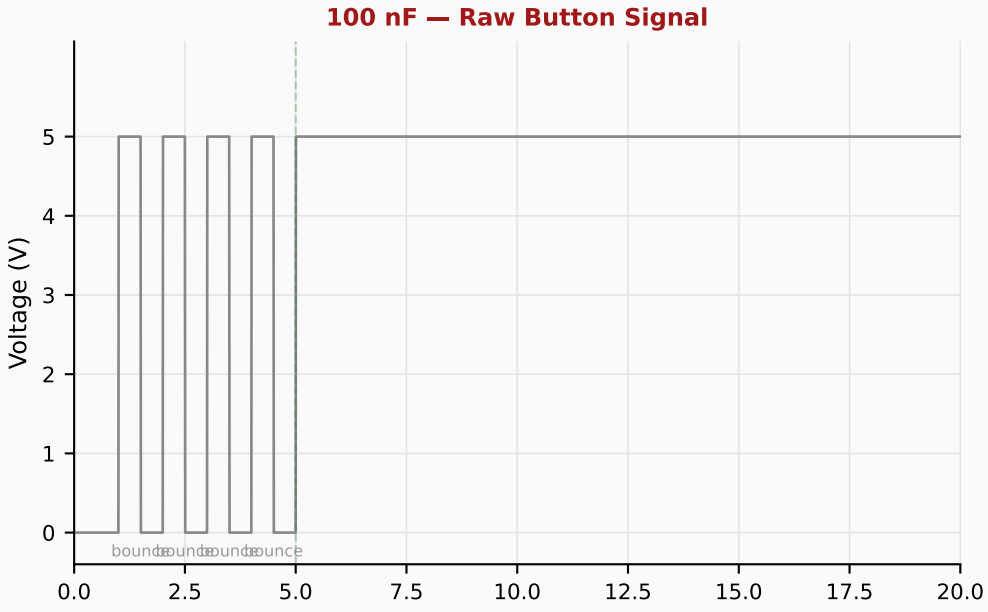
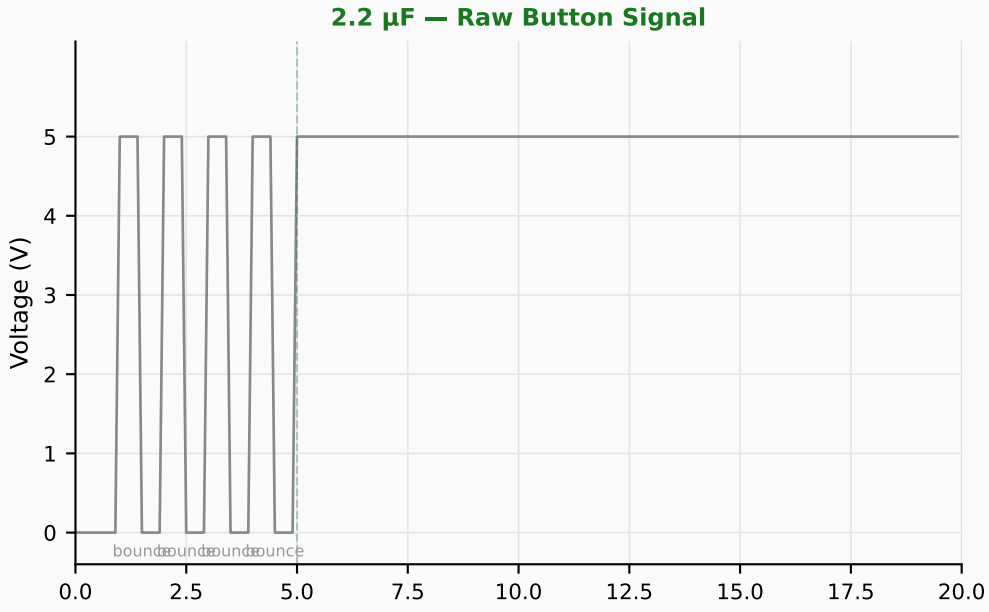
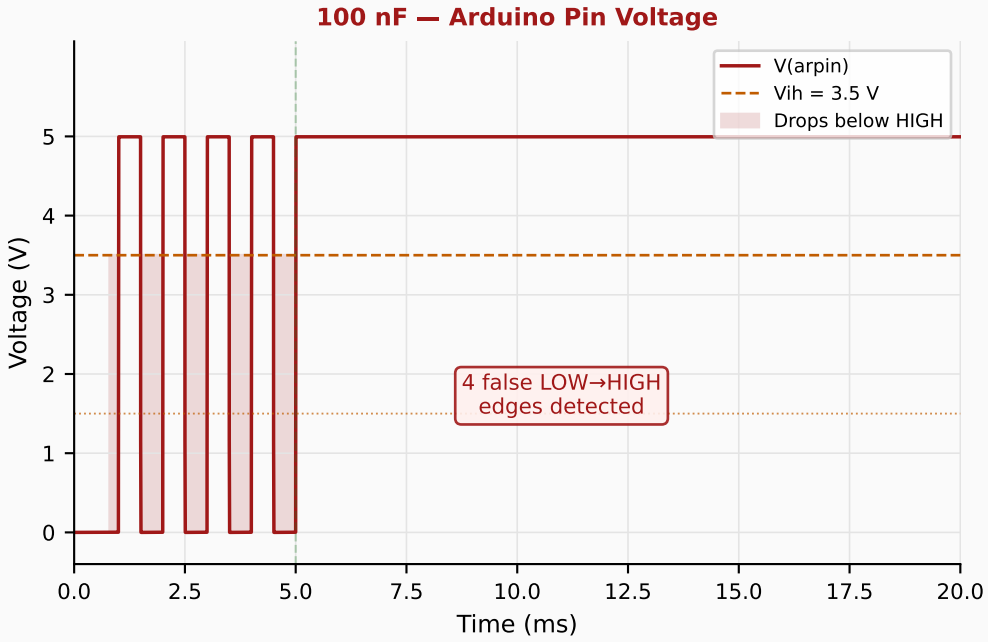
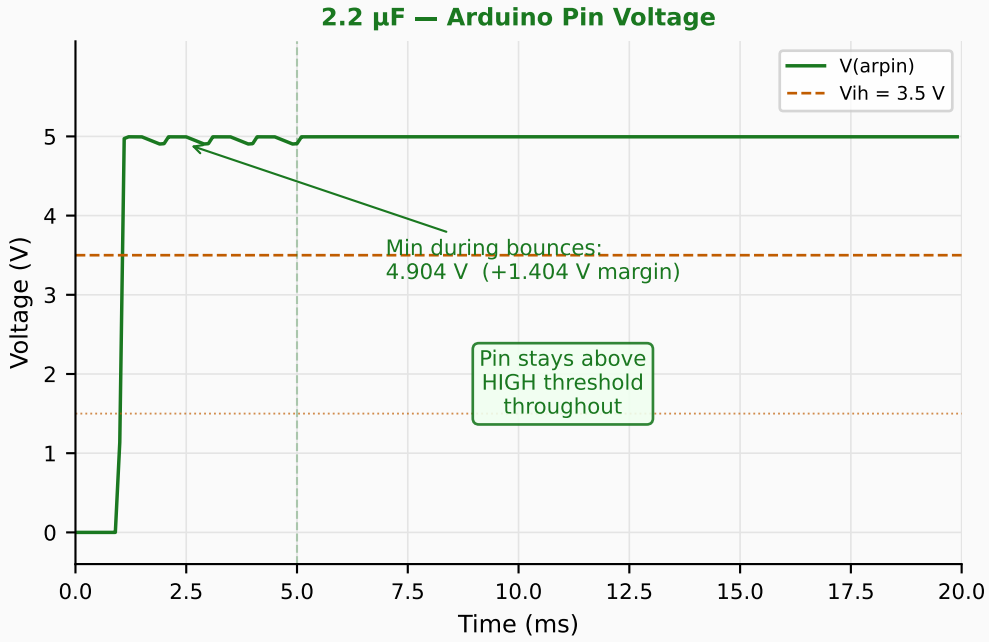
Button Debounce RC Filter — ARM_BTN / FIRE_BTN

Left: 100 nF — Rev B as-built (FAIL) Right: 2.2 µF — recommended fix (PASS)









Button Debounce — Parameters & Fix

Parameter	Value	Notes
Filter cap (Rev B)	**100 nF**	RC τ = 1.0 ms — FAIL: pin drops below 3.5 V on every bounce
Filter cap (fix)	**2.2 µF**	RC τ = 22 ms — PASS: pin drops only 0.11 V during 0.5 ms boun
Pulldown resistor	**10 kΩ**	Unchanged — sets τ with cap; limits current draw
Discharge τ (100 nF)	**1.0 ms**	10 kΩ × 100 nF — bounce hold above 3.5 V = only 0.36 ms
Discharge τ (2.2 µF)	**22 ms**	10 kΩ × 2.2 µF — bounce hold above 3.5 V >> 20 ms
Typical bounce duration	**1 – 20 ms**	Tactile switches — τ must exceed worst-case bounce
Voltage drop / bounce	**0.11 V**	2.2 µF: 5 × (1 − e^(−0.5ms/22ms)) — pin stays at 4.89 V
ATmega328 Vih	**3.5 V**	0.7 × Vcc — Schmitt trigger HIGH threshold at 5 V supply
ATmega328 Vil	**1.5 V**	0.3 × Vcc — Schmitt trigger LOW threshold
Glitches (100 nF)	**4 edges**	Arduino sees 4 false button presses per physical press
Glitches (2.2 µF)	**0 edges**	No threshold crossing during bounces
Firmware debounce	**Required**	Read → wait 20 ms → re-read; accept only if still HIGH

NOTE: button_debounce.cir has been updated to use a voltage-controlled switch model. Re-run it in LTspice GUI to get full SPICE confirmation of the 2.2 µF result. The 2.2 µF curve above is computed analytically from the correct RC model.